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ANALOG IC DESIGN LAB REPORT

BAND GAP REFERENCE CIRCUIT

INTRODUCTION

A Bandgap Reference (BGR) is a circuit used in analog and mixed-signal integrated circuits to generate a stable reference voltage that remains almost constant despite variations in:

- Temperature
- Supply voltage
- Process variations

The typical output voltage of a bandgap reference is around 1.2 V, which is close to the bandgap voltage of silicon.

OBJECTIVES

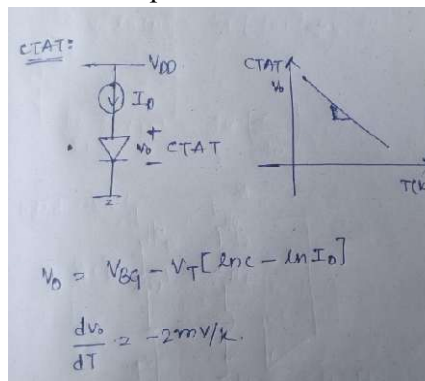
- To give a precise and stable voltage reference for accurate operation for the electronic circuits.
- To design a circuit that generates a temperature-independent reference voltage (~1.2 V).
- To achieve zero or minimal temperature coefficient by combining CTAT (V_{BE}) and PTAT voltages.
- To ensure the reference voltage is insensitive to supply voltage variations.
- To minimize the impact of process variations and ensure consistent performance across corners.

PRINCIPLE OF OPERATION

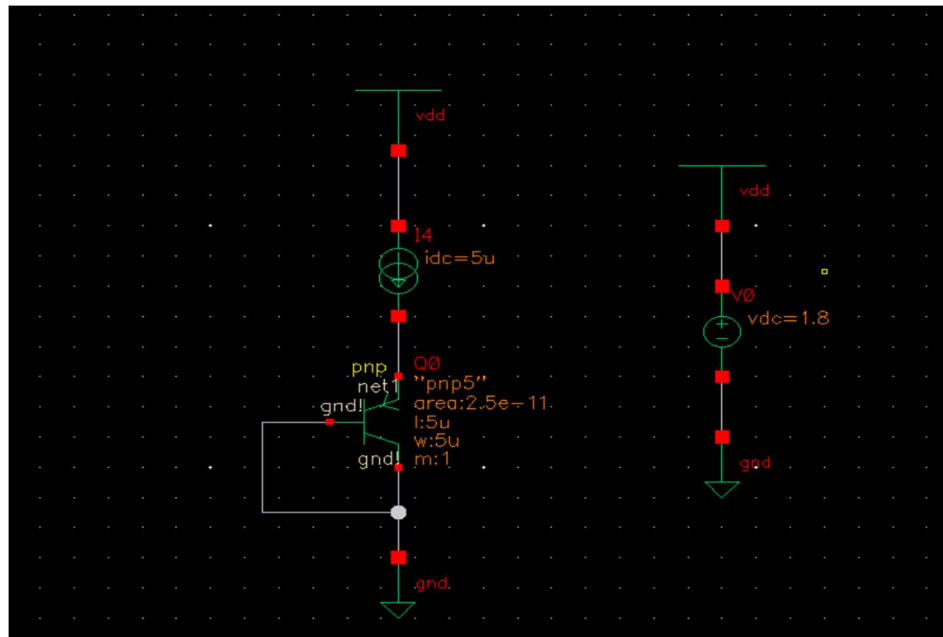
The bandgap reference works by combining:

1. CTAT Voltage (Complementary to Absolute Temperature)

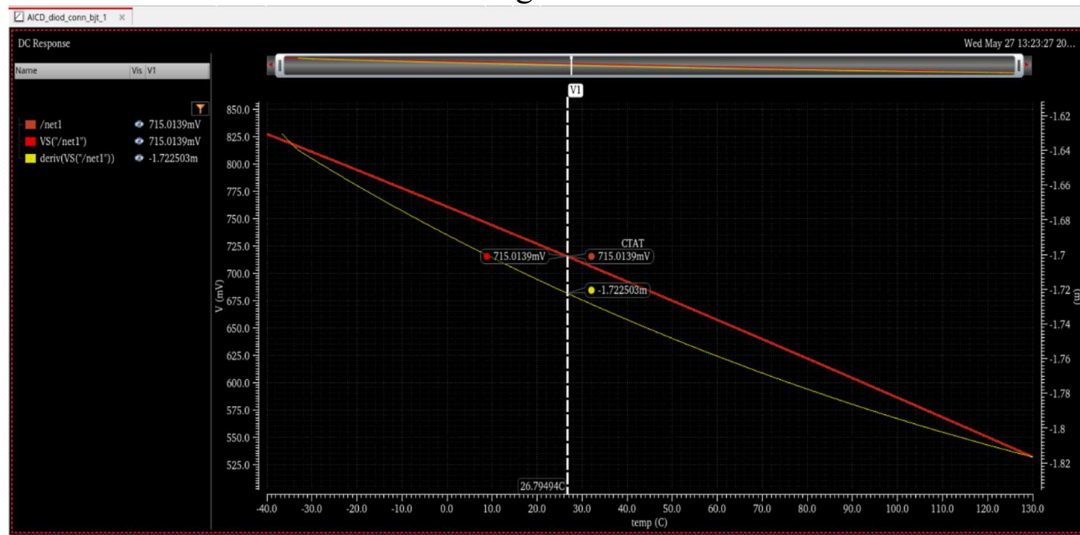
- This voltage decreases as temperature increases.
- Generated using a **forward-biased p-n junction (diode or BJT V_{BE})**
- V_{BE} decreases $\sim -2 \text{ mV/K}$ as temperature rises



Schematic: CTAT circuit



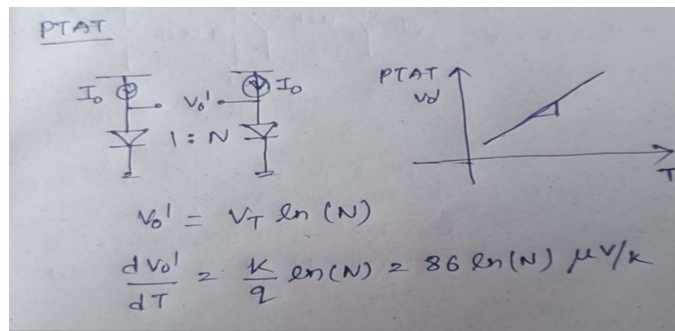
Simulation: CTAT behaviour made using a diode connected BJT & its SLOPE



Here, the CTAT from -40°C to 125°C is plotted (red). The yellow plot is the slope of the CTAT plot, where we got it as -1.72mV/K (approx -2mV/K).

2. PTAT Voltage (Proportional to Absolute Temperature)

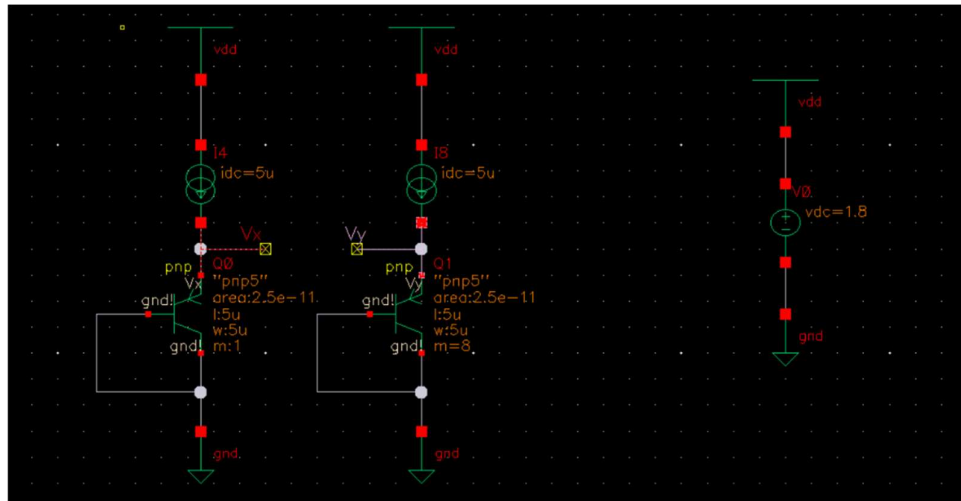
- This voltage increases as temperature increases.
- Generated using the difference between two transistor voltages operating at different current densities



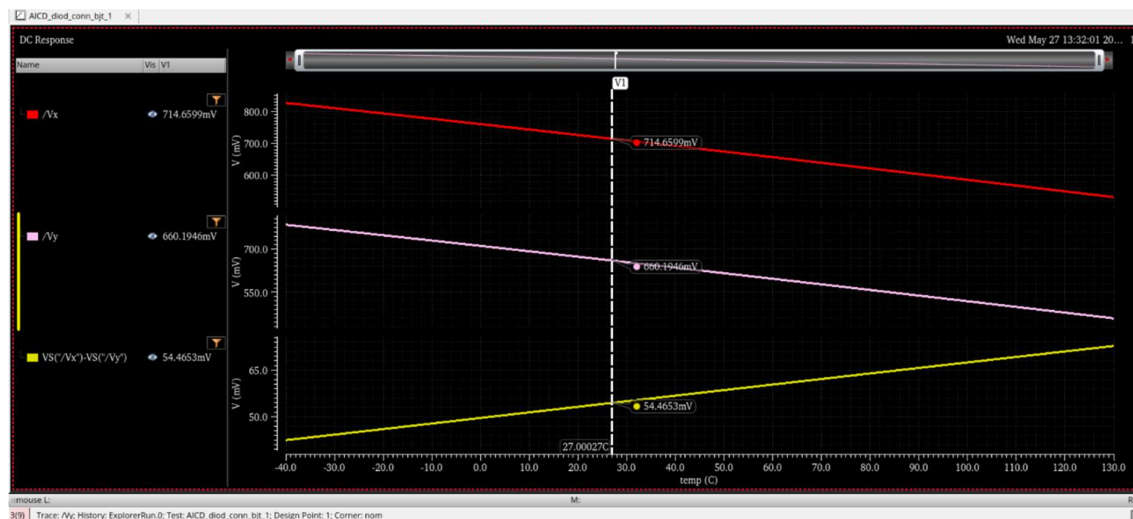
When CTAT and PTAT voltages are added properly, temperature variations cancel each other.

$$V_{\text{ref}} = \text{CTAT (vol)} + \text{PTAT (vol)} = V_{\text{BE}} + V_T \ln(n)$$

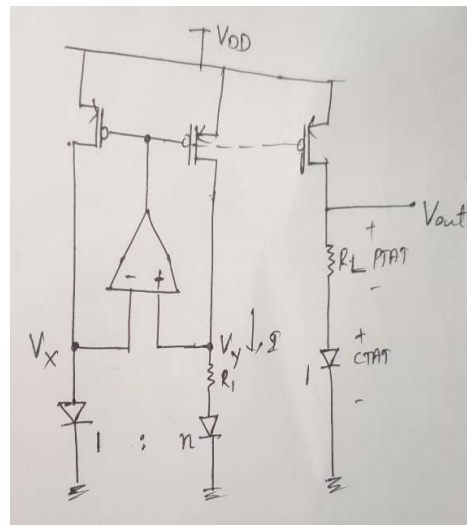
Schematic: PTAT circuit



Simulation: PTAT behaviour made using a diode connected BJT (yellow)



BASIC CIRCUIT DIAGRAM



Theory of Operation:

There is this need of having a constant bandgap voltage for the external circuits. So, we need bandgap Reference circuit which gives a constant voltage. BGR has two stable states — correct operating point AND a zero-current state (all transistors off). So, to maintain this, constant voltage all the time, we use a startup circuit (which consists of transistors). Startup circuit detects zero-current state and injects a kickstart current so to maintain the V_{ref} .

Here, we have made the constant voltage of $V_{ref} = 1.22V$ with a deviation of 10%.

- Identical PMOS transistors are used for controlled current sources which are used to make the currents in the branches equal.
- So here, we have used the Diodes with ratios 1: n(n=8) to generate CTAT voltage of ratio 1 and PTAT voltage of ratio 8 and by selecting the resistor and branch current of $I=5\mu A$ (in VY branch).
- Thus, here $R1$ is calculated with branch current I we have chosen and with these following relations, we also find out the $R2$, to increase the PTAT voltage, first we convert the voltage to current and thus by passing that current through a large resistor($R2$).

$$\begin{aligned}
 Q1: V_{EB1} &= V_T \ln \left(\frac{I_{C1}}{I_{S1}} \right) \\
 Q2: V_{EB2} &= V_T \ln \left(\frac{I_{C2}}{I_{S2}} \right) \\
 \text{As } I_{S2} &= n I_{S1} \\
 I &= \frac{V_{EB1} - V_{EB2}}{R1} \quad [I_{C1} = I_{C2} \text{ (opamp)}] \\
 I &= \frac{V_T \left[\ln \left(\frac{I_{C1}}{I_{S1}} \right) - \ln \left(\frac{I_{C2}}{I_{S2}} \right) \right]}{R1} \\
 I &= \frac{V_T \left[\ln \left(\frac{I_{C1}}{I_{S1}} \times \frac{n I_{S1}}{I_{C2}} \right) \right]}{R1} \\
 I &= \frac{V_T \ln n}{R1} \quad \Rightarrow V_T \ln n = I R1 \\
 \text{multiplier in BJT} & \quad \frac{V_T}{V}
 \end{aligned}$$

$$\begin{aligned}
 \text{Differentiate w.r.to temperature: } (n=8, n \text{ of multiplier in BJT}) \quad \frac{V_T}{V} \\
 \text{Prop } \frac{d}{dT} (V_T \ln n) &= \ln n \cdot \frac{dV_T}{dT} = \ln(8) \times \frac{K}{q} = \ln(8) \times 86 \mu V / ^\circ C = 178.83 \mu V / ^\circ C \\
 \text{CTAT } \frac{dV_{ref}}{dT} &= -1.72 \text{ mV} / ^\circ C \quad [\text{from graph}] \\
 I_{R1} &= I_{R1} + I_{R2} \Rightarrow V_{REF} = R1 (I_{R1} + I_{R2}) \\
 \therefore V_{REF} &= \frac{R1}{R2} \left[\frac{R1}{R1} V_T \ln n + V_{BE} \right] \quad \text{we need } \frac{dV_{REF}}{dT} \approx 0 \\
 \therefore \frac{R2}{R1} \times 178.83 \mu V / ^\circ C &= 1.72 \text{ mV} / ^\circ C \Rightarrow \frac{R2}{R1} = 9.619 \\
 \text{As we know, } \frac{V_T \ln n}{R1} &= I \Rightarrow R1 = \frac{26 \times 10^{-3} \times \ln(8)}{5 \mu A} \approx 10.81 K \\
 \therefore \text{Branch Current} &= 5 \mu A \quad \& \quad R2 \approx 104 K \\
 \text{of PTAT branch} &
 \end{aligned}$$

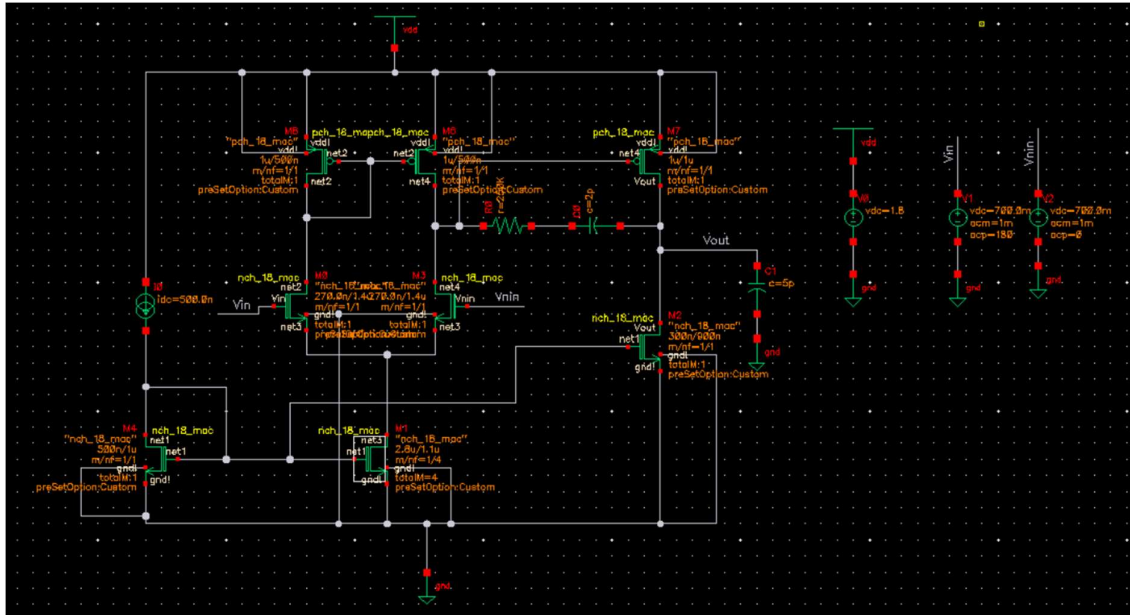
OP_AMP Design:

Specifications:

Gain $\geq 65\text{dB}$

Phase Margin $\geq 65^\circ$

We have chosen $I=0.5\mu\text{A}$, for the low power supply current specification of $I=10\mu\text{A}-20\mu\text{A}$.



Assumptions:

GBW = 1.0MHz

of PTAT branch

$\textcircled{*} \text{GBW} = 1.0\text{MHz}, C_c = 2\text{pF}, C_L \text{ taken} = 5\text{pF} \Rightarrow \text{for two stage, } C_c \geq 0.22 C_L$
 $C_c \geq 0.22 (5\text{pF})$
 $C_c \geq 1.1\text{pF}$

$g_{m1} = 2\pi \times 1.0\text{MHz} \times C_c = 12.56 \mu\text{S}$
 $\therefore \left(\frac{W}{L}\right)_0 = \left(\frac{W}{L}\right)_3 = \frac{(g_{m1})^2}{\mu_n C_{ox} \times I_{\text{into diff Pair}}} = \frac{(12.56)^2 \times 10^{-12}}{381.10 \times 10^{-4} \times 2.076 \mu\text{A}}$
 $\approx 0.1995 = \left[\text{chosen } \frac{W}{L} = \frac{270\text{nm}}{1400\text{nm}} \Rightarrow \therefore \frac{W}{L} = 0.1928 \right]$

As we have taken $I=0.5\mu\text{A}$ and to keep the **M4** in saturation, tuned the $(W/L)_4$ and set it to '0.5'. And now to get a tail current of '2uA', tuned the $(W/L)_1=2.5$ i.e., '0.5 $\times$ 4=2 (approx.)'. To keep the PMOS current mirrors in saturation, **M6** and **M8**, $(W/L)_6=(W/L)_8=2$ is fixed.

The W/L s of CS stage (stage 2) is kept in such a way that, they were in saturation. i.e., $(W/L)_7=1$ and $(W/L)_2=0.33$.

Parameters fixed:

$(W/L)_0 = (W/L)_3 = 0.5$

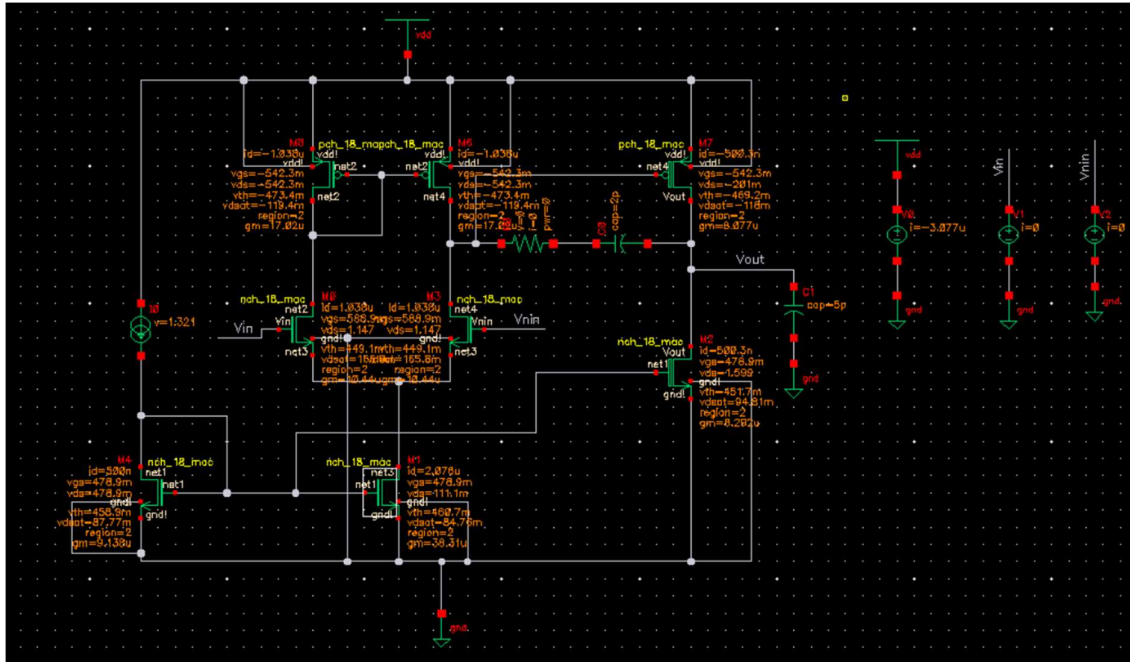
$(W/L)_1 = 2.54$

$(W/L)_2 = 0.33$

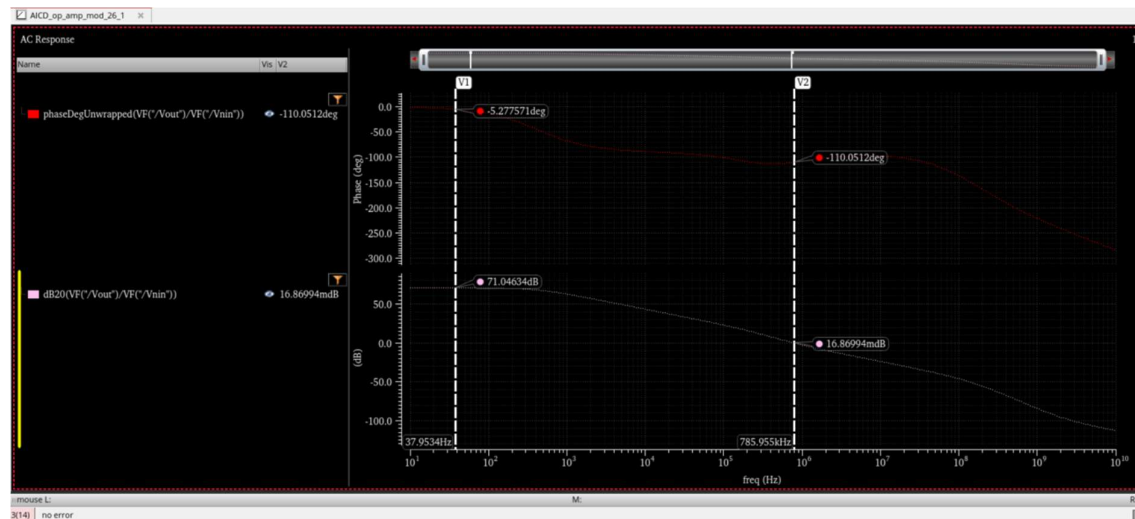
$(W/L)_4 = 0.5$

$(W/L)_6 = (W/L)_8 = 2$

$(W/L)_7 = 1$

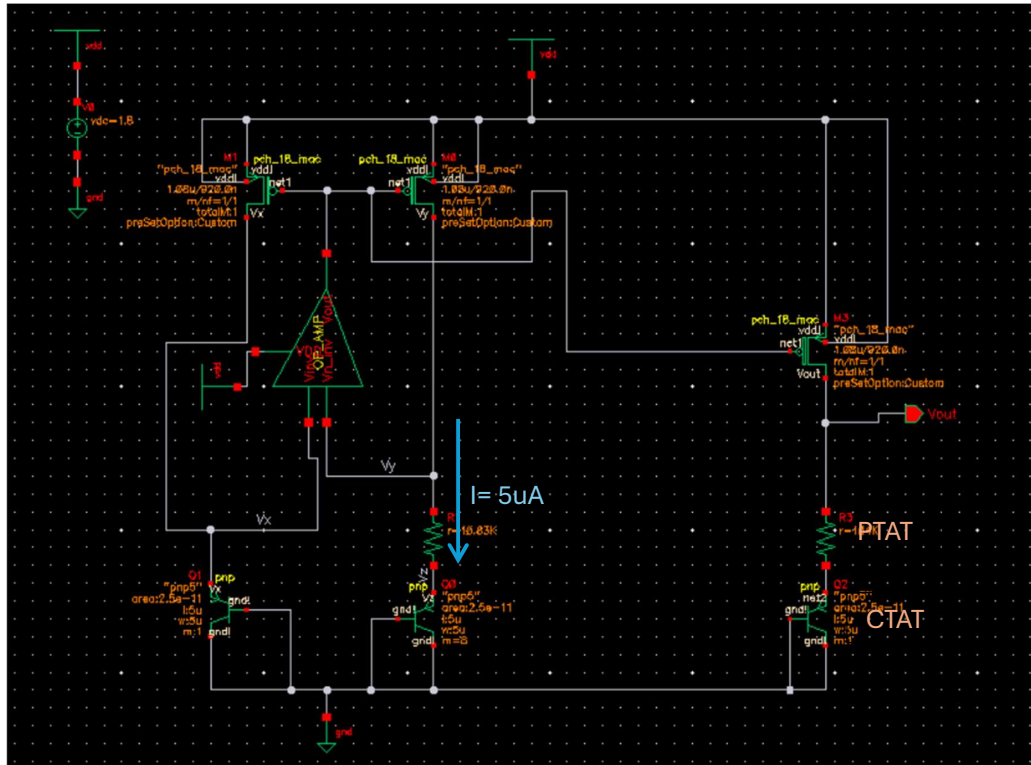


Gain & PM:



Gain: 71dB; Phase Margin: $180^\circ - 110.0521^\circ = 69.94^\circ$

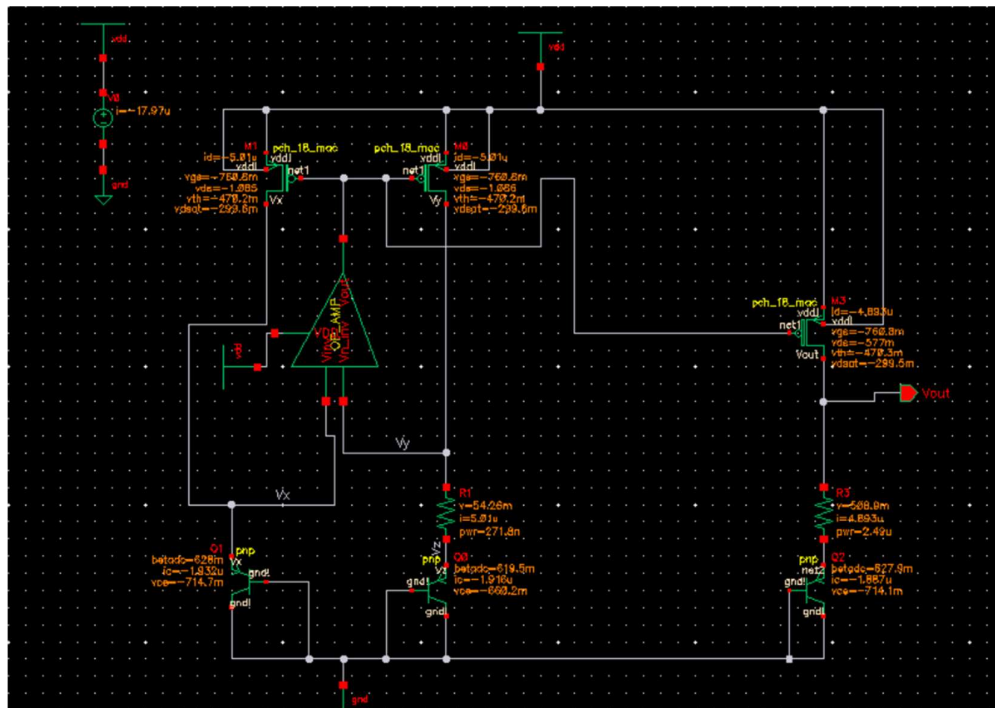
BGR circuit Schematic:



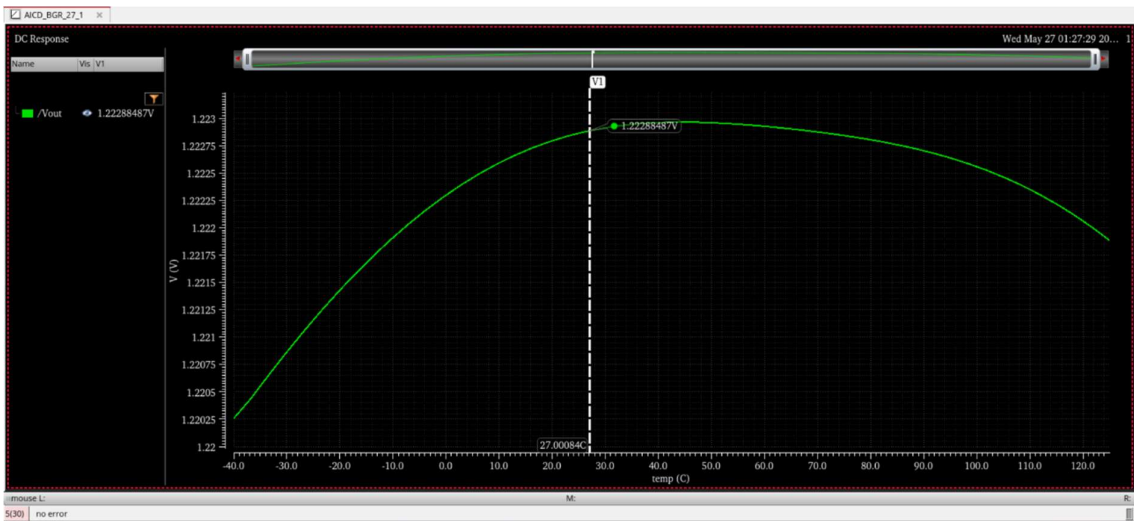
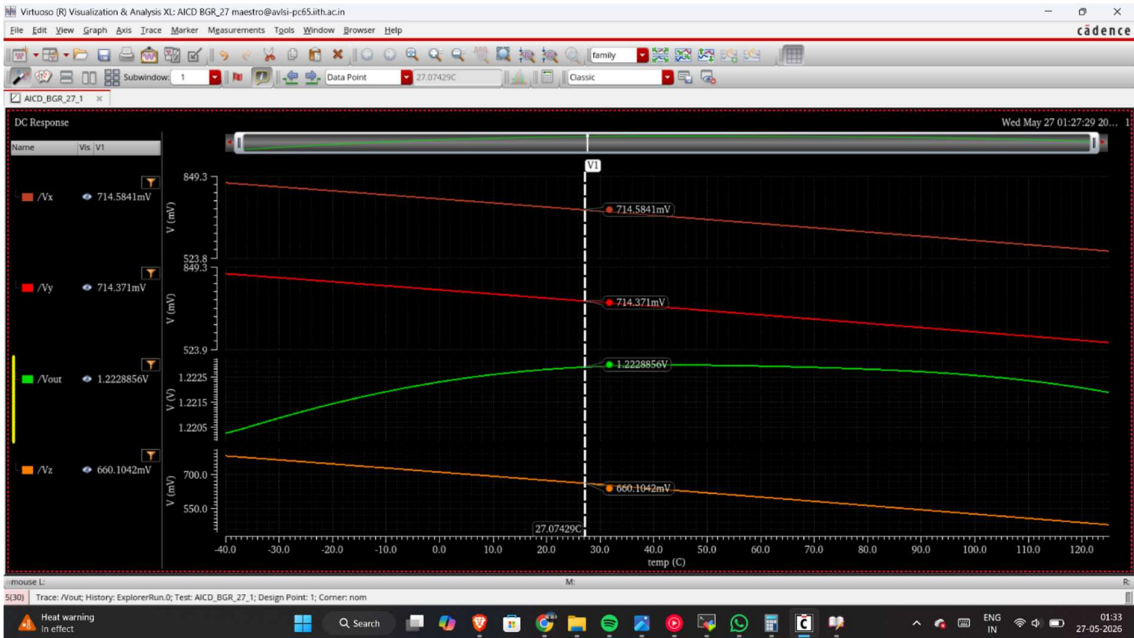
$Q1$ is the CTAT voltage diode connected BJT.

$Q2$ is the PTAT voltage diode connected BJT with $n=8$ maintaining the branch current of $5\mu A$ and the resistors $R1$, $R2$ and W/L s are selected accordingly. The supply current is in the required range i.e., $I=17\mu A$

V_x and V_y are the two inputs of op amp and constantly the op amp tries to make them equal and $V_P = V_x - V_y$, where the difference is amplified and gives us the required voltage of $1.22V$.

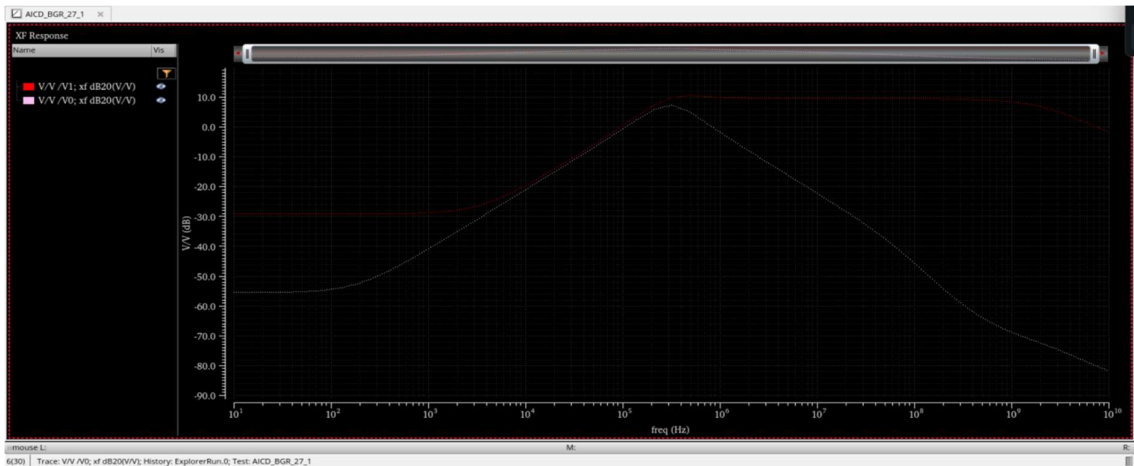


BGR circuit Simulation:



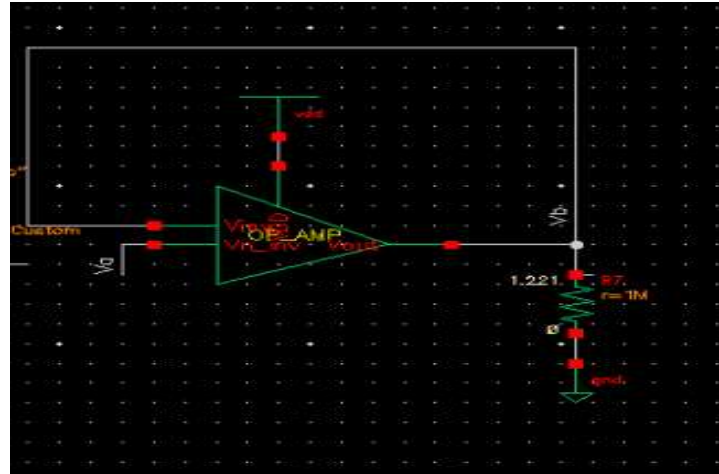
The output is maintained in the range of 1.2-1.25 which is called as $V_{ref}=1.22V$

PSRR plot:



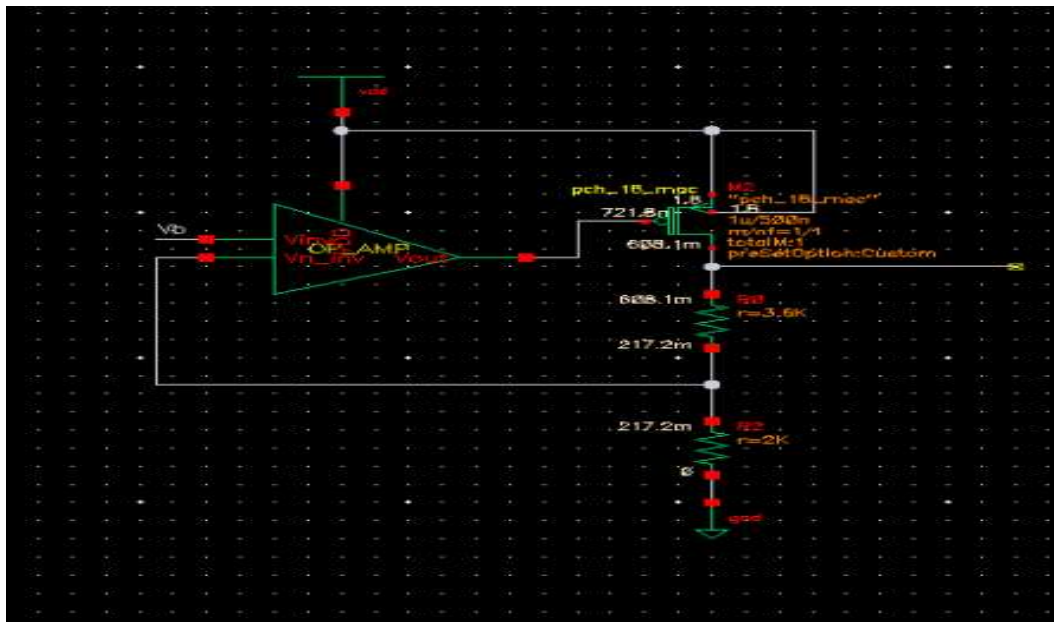
We have plotted the PSRR plot and got it approximately as 90dB.

Buffer schematic:



Buffer is used to reduce the loading effect on V_{2I} so that exact 1.22v will be given as input to V_{2I} .

V_{2I} schematic:



Here there observed a constant current in the pmos transistor, and we also know the voltage to be 0.6V, so by using ohm's law the required resistance is calculated to be $R_1 + R_2 = 5.6K$ ohms. And we have reduced the 1.22v to 0.6v by using the V_{2I} converter module.

The diagram illustrates a three-stage operational amplifier circuit. The first stage is a differential pair with a tail current source, followed by a common-emitter stage. The second stage is a differential pair with a tail current source. The third stage is a common-emitter stage. The output is taken from the collector of the third stage. The circuit includes various components like resistors, capacitors, and op-amp blocks.

DC Response

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Output of BGR

Output Of VZL

27.4640°C

1.222037V

607.6522mV

temp (C)

APPLICATIONS

It is widely used in:

- Analog-to-Digital Converters (ADCs)
- Digital-to-Analog Converters (DACs)
- Voltage regulators
- PLLs
- Sensors
- Power management circuits

CONCLUSION

A Bandgap Reference is an essential analog building block used to generate a stable reference voltage independent of temperature and supply variations. By combining PTAT and CTAT voltages, the circuit achieves excellent stability, making it highly suitable for modern integrated circuits and VLSI applications.